

Study tools for students

What Turbo AI does, what the research says works, and how Smart Learning can do it better for Jamaican students

Research and blueprint. Prepared 10 October 2026, revised the same day: the source for study material is the lesson the teacher already built in Smart Learning, so no upload is needed for it. This belongs to the Learning product only (assessments, results and report cards are not touched).

1. The short version

What Turbo AI is. A consumer study app, formerly called TurboLearn. A student uploads a PDF, a YouTube link or a lecture recording and Turbo produces notes, flashcards, multiple choice quizzes with explanations, a podcast-style audio summary, and a chat box that answers questions about those notes. It runs on web, iOS and Android, has a free tier and a paid tier of roughly \$10 to \$20 a month, and its terms say users must be 16 or older.

What we found.

- **The idea is sound, the product is thin.** Turbo is a clean pipeline from "my material" to practice. Its "AI tutor" is a chatbot over your own notes. It is not a structured tutor.
- **The research favours the quiz and flashcard half, not the notes half.** Practice testing and spaced practice are the two best-supported study methods; summarising and rereading are among the weakest. Generated notes are the least valuable thing Turbo makes.
- **Unguarded AI help can lower exam results.** In a randomised trial of nearly a thousand high school maths students, those given a plain ChatGPT-style helper did better in practice and worse on the later exam than students with no AI at all. A tutor built to give hints removed most of that harm.
- **Smart Learning already has most of the hard parts.** A hint-first AI tutor with safety flags and school consent, spaced-repetition flashcards that work offline, three-level lesson checks, topic tracking and a library. What is missing is the step that turns a finished lesson into study material for students (flashcards, practice questions and a short guide) without the teacher doing any extra work.

What we would build, in order.

1. **Lesson study guides:** when a teacher publishes a lesson, Smart Learning drafts a study guide from it: the key terms as flashcards, practice questions at Support, Core and Stretch levels with explanations, and a short list of key points. The teacher checks it once. Nothing is uploaded, because the lesson is already in Smart Learning.
2. **Ask your lesson:** the tutor that already answers questions about a lesson, now also able to quiz the student and draw on the guide, still hint-first.
3. **Today's practice:** one daily set that mixes due flashcards, weak topics from My Topics and new questions, so spaced practice happens by default.
4. **Your own notes:** later, a student can add their own notes (a PDF or a photo of handwritten notes) for subjects or material that are not in a lesson, and get the same flashcards and quiz.

What we would leave out. Live lecture recording (consent and privacy), "brainrot" video gimmicks, public note sharing, cross-app tracking, and scraping YouTube.

Cost. A lesson guide is made once per lesson and shared by the whole class, so it costs about 9 cents per lesson on Claude Sonnet 5.5 (under half a cent on Claude Haiku 5.5), however many students use it. The ongoing cost is tutor messages: about half a cent each on Sonnet. These are estimates to be measured in the pilot (section 9).

Decisions needed from you are in section 12.

2. What Turbo AI is and what it actually does

Everything below is taken from Turbo's own site, its blog and terms of service, and a small number of third-party reviews. Where the only source is Turbo itself, that is stated. Turbo's marketing numbers (millions of users, "3x faster", "99% accuracy") could not be verified and are not used in this blueprint.

Feature	What Turbo says it does	Evidence	Worth taking?
Notes from uploads	PDFs, YouTube links, audio and live recordings become organised notes in a Google Docs-style editor, in nested folders that sync across web, phone and tablet.	Turbo's site and blog	Partly. Useful as a reading aid and as the base for practice, not as a study method on its own.
Flashcards	Generated automatically for key concepts; "smart spaced repetition adapts to your pace"; Learn Mode sorts cards into New, Learning and Mastered.	Turbo's site and blog	Yes. We already have the spacing engine; we need the generation.
Quizzes	Multiple choice from your notes; choose topics or chapters; adjustable difficulty from basic to exam level; an explanation for every answer.	Turbo's site	Yes. The most valuable feature.
Chat with your notes	A chatbot inside the notes that answers questions without pasting anything and "answers at the level of the material".	Turbo's blog; no detail on citations	Yes, with citations and hint-first behaviour.
Podcast audio	Notes turned into a quick summary or deep dive audio, downloadable for offline listening.	Turbo's site	Later. Helpful on long bus rides; needs a speech service.
Lecture recording and notetaker	Records a live lecture, transcribes it and makes notes.	Turbo's site; reviewers report failed audio uploads	No. Consent, privacy and classroom policy problems outweigh the value.
Book library	Browse thousands of academic books, read, highlight and generate notes and flashcards from any chapter.	Turbo's site	Yes, later, through the Smart Learning Library once it has books.
Collaboration	Share notes with classmates, live editing, inline comments.	Turbo's site	Teacher-controlled class sharing only. No open sharing of student uploads.

Feature	What Turbo says it does	Evidence	Worth taking?
STEM formatting	Keeps maths formulas, chemical equations and code readable.	A review; Turbo's own blog admits its flashcards "didn't always include equations"	Essential for maths and science. We already render maths.
Premade study guides	A library of AP study guides.	Turbo's site	Our equivalent is curriculum-aligned lessons and packs built on the National Standards Curriculum.

Pricing, as far as it can be established

Turbo's site gives no prices. Third-party listings disagree: a free tier with tight monthly limits (one listing says about 2 hours of lecture processing, 1 PDF, 5 quiz sessions and 10 chat messages), and a paid tier between \$5.99 and \$49 a month, most often quoted at \$9.99 a month billed yearly or \$19.99 monthly. Treat all of these as rough.

What its terms and reviews tell us

- Minimum age 16; 16 and 17 year olds need a parent or guardian's consent. Our students are mostly 11 to 18, so Turbo itself would not be suitable for most of them without a school arrangement.
- Uploads are processed by third-party AI providers it names: Grok, Claude, OpenAI, Anthropic, Together AI, Google and Modal Labs. It may change them at any time.
- It says it is "not the system of record" for your files and makes no promise to keep inputs. It does not clearly say whether your content trains AI.
- Users are "solely responsible" for having the right to upload, and are told not to upload health, financial or ID information.
- Outputs may contain errors and "hallucinations" and users must review them. A third-party review found summaries sometimes omit details; another reports bugs, slow support and users feeling pressed to subscribe to keep their notes.
- The same review says users are tracked across other apps and sites by user and device ID. That is not acceptable for a school product.

What could not be verified

- User numbers, "3x faster" learning and any accuracy percentage: no method or source given.
- Exact free and paid limits, supported languages, and how the spaced repetition schedule actually works.
- Whether its "AI tutor" does anything beyond answering questions about your notes. Nothing found suggests it teaches in steps or adapts a learning path.

3. What the research says works

Before choosing features it is worth knowing which study behaviours actually help, and where AI helps or harms.

Finding	What it showed	What it means for us
Practice testing and distributed practice are the best-supported study techniques	A review of ten common techniques (Dunlosky and colleagues, 2013) rated practice testing and spacing as high utility across ages. Summarising, highlighting and rereading were rated low, although students use them most.	Put quizzes, flashcards and spacing at the centre. Treat generated notes as the starting point for practice, not the product.
Unrestricted AI help can hurt later performance	In a randomised trial with nearly 1,000 high school maths students (Bastani and colleagues, PNAS 2025), a plain GPT-4 helper improved practice scores but students scored 17% lower on the later exam than students who never had it. A tutor built to give hints largely removed the harm.	The tutor must give hints before answers and ask the student to try first. Our existing tutor already does this; every new feature must keep it.
Students misjudge their own learning	In the same study students with the unrestricted helper believed they had learned as much as their peers while doing worse on the exam.	Show students what they can actually do (quiz results, topics that need work), not how confident they feel.

Practical rules that follow: default to testing rather than summarising; make the student attempt before they see an answer; show them their own gaps; and make answers traceable to the source so wrong ones can be caught.

4. Where Smart Learning is today

Capability	Smart Learning now	Turbo AI	Gap
Spaced repetition flashcards	Yes. Leitner boxes (back after 0, 1, 3, 7 and 21 days), study sessions, offline use. Students make their own decks.	Yes	We need AI to make the cards
AI tutor	Yes. Asks about one teacher lesson, hints before answers, safety flags that an adult reviews, school consent, rate limits, blocked during exams.	Chat over notes	Ours is safer; it is limited to teacher lessons
Practice questions	Yes. Lesson checks at Support, Core and Stretch levels, AI-drafted by teachers.	Yes	Students cannot make their own
Topic and progress tracking	Yes. My Topics (from marked results), My Progress, catch-up for absent students.	Weak areas	None for tracking; needs practice results added
Library and audio	Library built, catalogue empty; read-aloud button on content.	Book library, podcasts	Needs books; audio summaries missing
Study guide, flashcards and quiz made from a lesson	No. Teachers can already draft a student version of a lesson and AI-draft check questions, but there are no lesson flashcards and no guide.	Yes, from uploads	Main gap, and the cheapest to close because the lesson already exists
Upload own notes	No	Yes	Useful later, for material outside lessons
Answers that cite the source	Lesson-bound, no page citations	Not described	Add

Capability	Smart Learning now	Turbo AI	Gap
Photo of handwritten notes	No	Not described	Add (strong fit for our students)

What we have that Turbo does not

The National Standards Curriculum, 5E and depth-of-knowledge alignment; teachers in the loop; school consent and oversight; flagged conversations for adults; an exam lock; offline and low-data use; and one login with Smart Assess and the school's topic list.

5. Principles the build must follow

1. **Testing before telling.** The default output is questions and flashcards, not summaries.
2. **Always grounded.** Every card, question and answer points to the page or passage it came from. Nothing is shown without a source.
3. **Hints before answers.** The student attempts first; the tutor guides. This is already the tutor's rule and is non-negotiable.
4. **A teacher can see and shape it.** Teachers can create packs for their class, review what students are told, and see flagged conversations. Students' private packs are not read by teachers without a flag.
5. **Works on a poor connection.** Generated sets can be used offline once downloaded; generation itself is a short online step with clear progress.
6. **Cost is capped per student.** Monthly allowances like the ones already used for AI drafting, so a class cannot run up an unexpected bill.
7. **Never in an exam.** Study tools and the tutor are unavailable while a student has an exam or test in progress.
8. **Learning only.** Practice results stay in Smart Learning. They never become marks, report card grades or Assess results.

6. The features

Size key: S is 1 to 2 days of build, M is 3 to 5 days, L is 1 to 2 weeks. Review, testing and a classroom pilot take longer than the build itself.

F1. Lesson study guides (build first)

Why this replaces "teacher uploads material". A lesson already holds the teacher's five steps (Engage, Explore, Explain, Elaborate, Evaluate), their key terms, resources, subject, grade and topic, and the teacher has already approved it. That is better source material than any uploaded file: it is the teacher's own wording, aligned to the curriculum, already checked, and there is no copyright or personal data question.

What the teacher does	Publishes a lesson as they do now. A study guide draft appears next to it. They look it over, fix or remove anything, and switch it on. That is the only extra step. A "draft all guides" button covers lessons they already have.
What the student sees	A "Study guide" tab on the lesson, after the lesson: a few key points in plain words; the key terms as flashcards they can add to their own deck (spacing works as it does today); practice questions at their suggested level that ask first and explain after; and "what you should be able to do".
What is generated	Key term cards (from the lesson's own key terms and the Explain step), 10 or so practice questions with explanations, five key points, and a "can you do this" list. Every item shows which step of the lesson it came from. Questions come first on the screen and the key points are short, because testing helps learning more than summaries do.
One guide per lesson	It is generated once and shared by the class. Students get different questions by level and by weak topic, but no one waits for generation and the cost does not grow with the number of students.
Reuses	The lesson and its steps, key terms and topic, the three check levels and the existing AI drafting of check questions, the flashcard decks and spacing engine, the teacher review pattern, topics and My Topics, offline use, the school consent switch.
New	A guide table, a draft step, a teacher review screen and the student guide tab.
Size	M
Main risks	A wrong card. Mitigated by the teacher check, source references to the lesson step, and a "report this" button.

F2. Ask your lesson

What the student does	Asks a question inside the lesson, as now, and can also say "quiz me". The tutor uses the lesson and the guide, quotes the step the answer comes from, and gives a hint before an answer on practice questions.
How	The tutor already exists with its rules, flags and consent. This adds the guide as extra context and a quiz mode.
Size	S to M

F3. Today's practice

What the student sees	One card on the Smart Learning home page: "12 cards due, 4 questions on Fractions, 3 new". A ten minute session mixes due flashcards, topics My Topics shows are weak, and new questions from the guides of lessons they have done.
How	Uses spacing dates already stored, weak topics already computed and the guide questions. No new AI cost.
Important	Practice results are recorded separately from marks and never feed report cards or Assess.
Size	M

F4. Your own notes (later)

What the student does	Adds their own notes as a PDF, a Word file or photos of handwritten notes, picks the subject, and gets flashcards and a quiz with the page each came from. A photo is read back to them to correct before anything is made.
Why later	It is the highest risk part (copyright, personal data, quality of photos) and the lessons already cover what teachers teach. It matters for material outside lessons, such as other subjects, notes taken in class and revision sheets.
Rules	Up to 30 pages or 10 MB; 10 packs a month (adjustable); private to the student; a warning and an automatic check for personal details; original deleted within 24 hours; the school can switch it off.
Size	L

F5. Teacher upload for material not in Smart Learning (small)

What it is	A teacher can add a worksheet, past paper notes or a textbook extract that is not in a lesson and make a guide from it, using the same pipeline as F4 with the teacher review of F1.
Size	S after F4

F6. Audio review

What the student does	Presses "Listen" on a guide and hears the key points read aloud.
How	Start with the speech built into the phone or browser (free, works offline, quality varies). Move to a paid speech service only if students use it and quality complaints justify the cost.
Size	S to M

F7. Library chapter to guide

What it is	Once the Smart Learning Library has books, a chapter can become a guide with the same pipeline.
Size	S after F4

What we will not build

Turbo feature	Why we leave it out
Live lecture recording and AI notetaker	Recording a classroom raises consent, privacy and school policy problems, and the learning value is lower than practice. Turbo's own terms push the responsibility onto students.
"Brainrot" and celebrity video explainers	Promotional gimmicks with no evidence of learning benefit.
Open sharing of notes	Student uploads can contain personal and copyrighted material. Sharing stays teacher-controlled and class-only.
YouTube link ingestion	Unreliable transcripts and platform terms. Teachers already add video links to the Videos page.

Turbo feature	Why we leave it out
Cross-app tracking and a paywall	Not appropriate for a school product.

7. How it is built

One pipeline, used by every feature

A lesson needs none of steps 1 to 3 below: its text is already in the database, already split into steps, already approved and already tagged with a topic. Only uploads, photos and library chapters go through them.

1. **Upload and check.** File type, size and page count; a scan for obvious personal data; the student's allowance.
2. **Read the text.** Use the PDF's own text where there is some (cheap and exact). Scanned pages and photos go through the vision model. Word files are converted. Page numbers are kept.
3. **Split into passages** with numbers and page references.
4. **Draft.** Ask the model for structured output: cards (front, back, passage) and questions (stem, options, answer, explanation, passage, level). The instruction is to use only the material and write maths in plain notation.
5. **Check.** The structure is valid; every cited passage exists; the key words of the answer appear in the cited passage, otherwise the item is dropped or marked "check this"; duplicates are removed.
6. **Save.** Cards go into the existing flashcard decks. Questions go into a new table. The pack stores where each item came from, not the file.
7. **Study.** Spacing, quiz attempts and a wrong-answer review.
8. **Clean up.** The original is deleted within 24 hours. Deleting a pack deletes its passages, cards and questions.

What is added to the database (proposed)

Table	Purpose
learning_study_packs	One row per guide or pack: where it came from (a lesson, an upload, a photo or a library chapter), the lesson it belongs to, owner, title, subject, grade, topic, status, and teacher approval. A lesson has one shared guide.
learning_study_passages	The extracted text split into numbered passages with page numbers. This is what answers are checked against.
learning_study_questions	Quiz questions with level, options, answer, explanation, source passages, hidden flag.
learning_study_attempts	A student's answers, used for practice and weak topics only. Never used for marks.
learning_study_reports	A student or teacher marking a card or question as wrong, with a reason.
learning_study_usage	Monthly counts and token use per student, for allowances and cost tracking.
Flashcard decks and cards (existing)	Gain a pack reference and source passages. The spacing engine and offline support are unchanged.
Tutor conversations (existing)	Gain an optional pack reference. Flagging, consent and review are unchanged.

Rules in the database

- A lesson guide follows the lesson: a student can open it only if they can open the lesson, and only after the teacher has switched it on. Students see only their own personal packs.
- A teacher sees the packs they made and class packs they assigned. Teachers do not browse students' private packs; flagged conversations and reported items reach the teacher or head of department as they do today.
- Only the owner can delete a pack, and deleting it removes everything derived from it.
- The school leadership switch that already governs the AI tutor also governs study packs. Nothing works until it is on.
- Rules like these are tested the way the rest of the project is: sandbox tests for every rule and a probe from a student account that tries to read, change and delete what it should not.

Models and limits

- Start with Claude Sonnet 5.5 for quality during the pilot; measure Claude Haiku 5.5 on the same 100 items. If accuracy is within tolerance, use Haiku to draft cards and questions and keep Sonnet for the tutor.
- Drafting runs in two steps (cards, then questions) with visible progress, so each call stays inside the 60 second limit of the hosting.
- Per student: 10 packs and 100 tutor messages a month by default, plus the burst limits already in place. All adjustable by the school.
- Failures use the same plain-language messages and alerts as the existing AI features, and never leave a half-made pack on screen.

8. Safety, privacy and rules

Concern	Rule
Age and consent	Turbo is for 16 and over. Ours is run through the school, so the school leadership approves the feature (as with the AI tutor) and informs parents. Students are mostly 11 to 18.
Personal data	Lesson guides contain no student data. For uploads: a clear warning, an automatic check for phone numbers, ID numbers and addresses, and the original file deleted within 24 hours. Prompts contain only the material.
Copyright	Lesson guides are made only from the school's own lessons. For the later upload phase: uploads are for the student's own studying, never shared with other students and deleted after reading, and the school should say in writing what students may upload.
Retention	Originals 24 hours. Packs until the student deletes them, with automatic clearing after 12 months unused (to be confirmed).
Who processes the data	Anthropic, through our existing AI connection. It must be named in the school's privacy notice. Confirm in Anthropic's current commercial terms that API data is not used for training. No other processor is added.
Accuracy	Every item is grounded in a passage and checked against it; students can report a wrong card; teachers review class packs; the pilot includes a teacher accuracy audit.
Over-reliance	Attempt first, hints before answers, and a tutor that does not do the work. Students are shown their own results, not just their confidence.
Exams and integrity	Study tools and the tutor are blocked while an exam or test is in progress. Live exam questions from Assess are never read into a pack.

Concern	Rule
Wellbeing and content	The existing flagging sends worrying messages to an adult; the tutor already declines unsuitable topics.
Cost abuse	Per student allowances, burst limits and a monthly cost report for the school admin.
Jamaican law	The Data Protection Act applies to children's personal data. Before the pilot, write a short data protection impact assessment and have the school's legal advisor confirm the wording of the notice and the consent letter.

9. What it costs to run

These use Anthropic's published prices (Sonnet 5.5: \$2 per million input tokens and \$10 per million output; Haiku 5.5: \$0.10 and \$0.50; cached reads far cheaper; a 50% discount for non-urgent batch work). The token counts are assumptions to be measured in the pilot, not measurements.

Operation	Assumed tokens	Sonnet 5.5	Haiku 5.5
A lesson study guide (once per lesson, shared by the class)	8,000 in, 4,000 out	about \$0.06	about \$0.003
600 lessons in a year (all guides)		about \$36 a year	about \$2 a year
One tutor message on a cached lesson	10,000 cached, 600 new, 300 out	about \$0.005	about \$0.0003
One student for a month (150 tutor messages)		about \$0.80	about \$0.05
Whole school, 30% of 1,800 students active		about \$420 a month	about \$25 a month
A student's own pack from 20 pages (later phase)	15,000 in, 6,000 out	about \$0.09	about \$0.005

- The monthly allowance caps one student, so the worst case is predictable. The lesson guides add almost nothing because they are made once, not per student.
- PDFs full of images cost many more tokens than text, so the pipeline reads the text layer first and only uses the vision model where needed.
- Caching and batch processing can reduce these further. The pilot will give real numbers before anything is switched on school-wide.

10. Build plan

Phase	What	Size	Done when
0. Decide and prepare	Your decisions (section 12), the wording for teachers, allowances, a short impact assessment, choice of pilot classes.	S	The decisions are made and the school leadership has approved the wording.
1. Lesson study guides	F1 and F2: a guide drafted from each published lesson, teacher check, the student guide tab, the tutor's quiz mode.	M	A teacher can switch a guide on; students can study and quiz; the accuracy audit passes; cost per guide is known.
Pilot	Two classes for four weeks, using guides on the lessons they already have.		The measures below are met.

Phase	What	Size	Done when
2. Today's practice	F3: the daily mixed set.	M	The set mixes spacing, weak topics and new questions.
3. Your own notes	F4 and F5: upload and photo, with the personal data check, allowances and the report button.	L	Students can make packs; safety tests pass.
4. Audio and library	F6 and F7.	S to M	Listening works offline; a library chapter becomes a guide.

How we will know it works

- Weekly active students and cards reviewed per student.
- Share of cards reaching box 4 or higher after four weeks (spacing is working).
- Quiz accuracy on the same topic two weeks later, and results on the teacher's next test against a similar class without packs.
- Teacher-flagged errors: fewer than 5 per 100 items and no serious ones.
- Tutor flags and how they were handled.
- Cost per active student against the estimate.
- A short student and teacher survey added to the existing survey.

Tests every phase must pass

- Teachers check 100 generated items: at least 95 correct and correctly cited.
- Prompts that try to get exam answers, personal information or harmful content are declined or flagged.
- A student with an exam in progress cannot open the tools.
- A student cannot read, change or delete another student's pack (sandbox tests and a live probe).
- Study works offline once a pack is downloaded.
- No uploaded original remains after 24 hours.

11. Risks

Risk	How it is handled
Wrong or misleading cards	Grounding and checking, a report button, teacher review of class packs, and an accuracy audit before wider use.
Students lean on the tutor and learn less	Attempt first, hints before answers, practice results shown to the student, and the exam block.
A data protection complaint	School approval and parent notice, minimal retention, no file kept, a named processor and a written impact assessment.
A copyright complaint	Lesson guides use only the school's own lessons. Later, uploads are private, never shared and deleted after reading.
Cost surprise	Allowances, burst limits, a monthly cost report and a cheaper model fallback.
Weak results on maths, diagrams or handwriting	Correction step for photos, plain notation for maths, and a pilot that tests these subjects first.
Teacher review becomes a chore	One quick check per lesson, a bulk "draft all" and approve with spot checks, and sensible defaults.

Risk	How it is handled
Scope creep into Assess	The product boundary rule: practice results never become marks or report card entries.
Dependence on one AI supplier	All AI calls already go through one wrapper, so a second supplier can be added without changing the features.

12. Decisions needed from you

	Decision	My recommendation
1	Build lesson study guides first, made from lessons already in Smart Learning?	Yes. It needs no upload and is the safest, cheapest and quickest.
2	Must a teacher approve each guide before students see it?	Yes, once per lesson, with a "draft all" button and a quick review screen.
3	Should the guide be switched on by default for new lessons?	Drafted by default, shown to students only after the teacher switches it on.
4	Monthly tutor allowance per student?	100 messages, adjustable by the school.
5	Student uploads of their own notes: when?	After the pilot, in two classes first.
6	Who approves the feature for the school?	The same leadership switch as the AI tutor.
7	Which AI model to start with?	Sonnet 5.5, with Haiku 5.5 tested on 100 items.
8	Pilot classes?	The Grade 9 mathematics class used for the demo data, plus one English class.
9	Audio?	Phone or browser speech first.
10	Name in the product?	"Study guide" on each lesson, under Smart Learning.

13. Sources

Turbo AI (vendor material).

- [1] Turbo AI home: <https://www.turbo.ai/>
- [2] Turbo AI for students: <https://www.turbo.ai/for-students>
- [3] Turbo AI terms of service: <https://www.turbo.ai/terms-of-service>
- [4] Turbo AI vs ChatGPT (Turbo's own blog): <https://www.turbo.ai/blog/turbo-vs-chatgpt>
- [5] Turbo AI vs StudyFetch (Turbo's own blog): <https://www.turbo.ai/blog/studyfetch-vs-turbo>

Reviews.

- [6] HyScaler review (marketing in tone; pricing and tracking notes): <https://hyscaler.com/insights/discover-turbolearn-ai/>
- [7] Tools for Humans listing: <https://www.toolsforhumans.ai/ai-tools/turbolearn-ai>

Research.

- [8] Dunlosky and colleagues (2013), Improving students' learning with effective learning techniques: <https://doi.org/10.1177/1529100612453266>
- [9] Bastani and colleagues (2025), Generative AI without guardrails can harm learning, PNAS: <https://doi.org/10.1073/pnas.2422633122>
- [10] Wharton summary of the Bastani study: <https://knowledge.wharton.upenn.edu/article/without-guardrails-generative-ai-can-harm-education>

Prices.

[11] Anthropic pricing: <https://platform.claude.com/docs/en/about-claude/pricing>

Limits of this research. Turbo AI is a closed product: how its generation and spacing really work, its accuracy and its limits could not be tested directly without an account. Third-party reviews were of mixed quality and are flagged where used. Vendor claims are labelled as claims.